

STATISTICAL MODELLING SUPPORT GROUP

STATISTICAL MODELING

Multiple Linear Regression - Guided Practice

Step-by-Step Calculation Questions

Useful equations

$$\hat{\beta} = (X^T X)^{-1} X^T Y, \quad \hat{Y} = X \hat{\beta}, \quad e = Y - \hat{Y},$$

$$TSS = \sum (y_i - \bar{y})^2, \quad SSE = \sum (y_i - \hat{y}_i)^2, \quad R^2 = \frac{TSS - SSE}{TSS},$$

$$MSR = \frac{TSS - SSE}{k}, \quad MSE = \frac{SSE}{n - k - 1}, \quad F = \frac{MSR}{MSE}.$$

Work through the parts in the order given. Show the main calculation at each step and give a short interpretation where requested.

Question 01

A company models weekly sales Y , in thousands of units, using advertising expenditure X_1 , in thousands of rupees, and product price X_2 , in hundreds of rupees:

$$\hat{Y} = 18 + 2.5X_1 - 1.2X_2.$$

- Calculate the predicted sales when $X_1 = 8$ and $X_2 = 5$.
- Recalculate the prediction when advertising increases to $X_1 = 10$, while price remains 5.
- Find the change between the two predictions.
- Interpret the coefficients (2.5) and (-1.2), holding the other predictor constant.

Question 02

For three observations, the design matrix, observed response and estimated parameter vector are

$$X = \begin{pmatrix} 1 & 2 & 5 \\ 1 & 4 & 3 \\ 1 & 6 & 4 \end{pmatrix}, \quad Y = \begin{pmatrix} 10 \\ 14 \\ 20 \end{pmatrix}, \quad \hat{\beta} = \begin{pmatrix} 10 \\ 2 \\ -1 \end{pmatrix}.$$

- Identify the intercept column and the two predictor columns in (X) .
- Calculate $\hat{Y} = X \hat{\beta}$.
- Calculate the residual vector $e = Y - \hat{Y}$.
- Calculate $SSE = e^T e$.

Question 03

For a regression model with two predictors,

$$(X^T X)^{-1} = \frac{1}{10} \begin{pmatrix} 6 & -3 & 0 \\ -3 & 4 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \quad X^T Y = \begin{pmatrix} 50 \\ 80 \\ 60 \end{pmatrix}.$$

- Multiply the first row of $(X^T X)^{-1}$ by $X^T Y$ to obtain $\hat{\beta}_0$.
- Repeat the multiplication for the second and third rows to obtain $\hat{\beta}_1$ and $\hat{\beta}_2$.
- Write the fitted model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$.
- Predict Y when $x_1 = 2$ and $x_2 = 1$.

Question 04

Consider

$$X = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \text{ and } Y = \begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix}, \text{ and } (X^T X)^{-1} = \begin{pmatrix} 3 & -2 & -2 \\ -2 & 2 & 1 \\ -2 & 1 & 2 \end{pmatrix}.$$

- Calculate X^T .
- Calculate $X^T X$.
- Calculate $X^T Y$.
- Use the given inverse to calculate $\hat{\beta}$.
- Find $\hat{Y} = X\hat{\beta}$ and check whether any residual is non-zero.

Question 05

Teaching method has three categories: Classroom, Online and Hybrid. Classroom is the reference category. Let $D_1 = 1$ for Online and $D_2 = 1$ for Hybrid. A fitted model is

$$\hat{Y} = 50 + 4X + 6D_1 - 3D_2,$$

where (X) is weekly study hours.

- Write the dummy-variable pair (D_1, D_2) for each teaching method.
- Write a separate fitted equation for Classroom, Online and Hybrid.
- Calculate the predicted score for a student studying (5) hours under each method.
- State which method has the largest predicted score at ($X=5$).

Question 06

With Blue as the reference colour, a fitted model is

$$\hat{Y} = 30 + 2X + 8C_R + 4C_G - 2C_Y.$$

- Write the fitted equation for Blue, Red, Green and Yellow.
- At ($X=3$), calculate the prediction for each colour.
- Take Red as the new reference level. Find the new intercept by using the Red equation.
- Express the Green, Yellow and Blue effects as differences from Red.
- Write the re-expressed model and verify one prediction from part (b).

Question 07

For four observations,

$$Y = \begin{pmatrix} 6 \\ 8 \\ 9 \\ 12 \end{pmatrix}, \text{ and } \hat{Y} = \begin{pmatrix} 5 \\ 8 \\ 10 \\ 12 \end{pmatrix}.$$

- Calculate the response mean $\bar{y}$.
- Calculate every value of $(y_i - \bar{y})^2$ and hence obtain TSS .
- Calculate the residuals and hence obtain (SSE).
- Calculate R^2 .
- Interpret the calculated R^2 as a proportion of variation explained by the model.

Question 08

Two models are fitted to the same response data, for which ($TSS=500$). Model A has ($SSE=100$), while Model B has ($SSE=60$).

- Calculate the regression sum of squares ($TSS-SSE$) for each model.
- Calculate R^2 for each model.
- Calculate how much residual variation is reduced when moving from Model A to Model B.
- Based only on R^2 , identify the model that captures more of the observed response variation.

Question 09

A multiple-regression model has ($n=25$) observations, ($k=3$) predictors, ($TSS=480$), and ($SSE=120$).

- Calculate the regression sum of squares ($TSS-SSE$).
- Find the regression, error and total degrees of freedom.
- Calculate (MSR) and (MSE).
- Calculate the overall (F)-statistic.
- Arrange the results in a regression ANOVA table.

Question 10

An incomplete regression ANOVA summary is

Source	SS	df	MS
Regression	240	4	?
Error	?	30	8
Total	?	?	

- Calculate the regression mean square.
- Use the error mean square and error degrees of freedom to calculate (SSE).
- Calculate (TSS) and the total degrees of freedom.
- Determine the number of predictors (k) and observations (n).
- Calculate the (F)-statistic and complete the table.

Question 11

A regression model has $n = 40$, $k = 2$, $TSS = 900$, and $SSE = 360$. Use $F_{0.05,2,37} = 3.25$.

- State H_0 and H_1 for the overall regression test.
- Calculate the regression sum of squares, (MSR), and (MSE).
- Calculate the overall (F)-statistic.
- Compare the calculated statistic with the critical value and make the statistical decision.
- Explain why rejection means at least one slope is non-zero, rather than proving that both slopes are non-zero.

Question 12

A model fitted using $n = 30$ observations and $k = 3$ predictors has $TSS = 600$ and $R^2 = 0.75$.

- Rearrange the R^2 equation to calculate SSE .
- Calculate the regression sum of squares.
- Find the regression and error degrees of freedom.
- Calculate (MSR), (MSE), and (F).
- State what the value $R^2 = 0.75$ means in this model.

Question 13

A linear model gives the following fitted and observed values:

$\hat{y}_i$	2	4	6	8	10
y_i	4	4	5	8	12

- Calculate the five residuals $e_i = y_i - \hat{y}_i$.
- Calculate (SSE).
- Sketch the residuals against the fitted values.
- Identify the visible pattern and decide whether the linearity assumption appears reasonable.
- State how a polynomial term such as X^2 could help if the pattern is curved.

Question 14

A model has four predictors with the following VIF values:

Predictor	X_1	X_2	X_3	X_4
VIF	1.0	2.8	6.4	9.1

- Classify each value using: 1 = not correlated, 1 to 5 = moderately correlated, and greater than 5 = highly correlated.
- Identify the two predictors causing the greatest concern.
- Explain briefly why high correlation among predictors makes their separate effects difficult to determine.

Question 15

Residuals are grouped according to low, medium and high fitted values:

Fitted-value region	Residuals			
Low	-1	1	-0.5	0.5
Medium	-2	2	-1.5	1.5
High	-4	4	-3	3

- Find the residual range in each fitted-value region.
- Describe how the residual spread changes as the fitted value increases.
- Identify the assumption that is likely violated.
- Name the graphical pattern and one formal test from the lecture that could be used to investigate it.

Question 16

Three models produce Durbin–Watson statistics (0.6), (2.0), and (3.4).

- For each value, identify whether it suggests positive autocorrelation, no autocorrelation, or negative autocorrelation.
- Identify the model that best supports the independence assumption.
- State what can happen to estimated standard errors when residual errors are correlated.

Question 17

Model A has a Q–Q plot with clear departures from its reference line and a Shapiro–Wilk p-value of (0.018). Model B has points approximately following the reference line and a Shapiro–Wilk p-value of (0.31).

- State the null hypothesis assessed by the Shapiro–Wilk test.
- Make the normality decision for each model at the 5% level.
- Identify which model better satisfies the normality-of-residuals assumption.
- Explain why the diagnostics should be applied to residuals rather than only to the original response values.

Question 18

A fitted model has $R^2 = 0.91$ and a significant overall F -test. Further diagnostics give a largest VIF of 7.5, a funnel-shaped residual plot, $DW = 1.0$, and a Q–Q plot that is approximately linear.

- Interpret the R^2 value.
- State what the significant (F)-test establishes.
- Match the VIF, funnel shape, Durbin–Watson value and Q–Q plot to their corresponding assumptions.
- Identify which assumptions appear problematic and which appears satisfactory.
- Explain why the model should not be accepted using only R^2 and the F -test.

Question 19

To represent curvature, a model is fitted as

$$\hat{Y} = 12 + 3X - 0.5X^2 + 2Z.$$

- Treat X , X^2 , and Z as three predictor columns and write the corresponding row of the design matrix for $X = 4$ and $Z = 3$.
- Calculate the predicted response for ($X=4$) and ($Z=3$).
- Calculate the prediction after incorrectly omitting the X^2 term while keeping the other coefficients unchanged.
- Find the difference between the two predictions and explain the role of the polynomial term.

Question 20

A model with (n=20) observations and (k=2) predictors provides

$$(X^T X)^{-1} = \frac{1}{10} \begin{pmatrix} 6 & -3 & 0 \\ -3 & 4 & -1 \\ 0 & -1 & 2 \end{pmatrix}, \text{ and } X^T Y = \begin{pmatrix} 50 \\ 80 \\ 60 \end{pmatrix},$$

together with $TSS = 1000$, $SSE = 250$, VIF values 1.8 and 6.2, $DW = 1.2$, a funnel-shaped residual plot, and a Q-Q plot that is approximately linear. Use $F_{0.05,2,17} = 3.59$.

- a) Calculate $\hat{\beta}$ and write the fitted regression equation.
- b) Calculate R^2 .
- c) Calculate the regression sum of squares, degrees of freedom, (MSR), (MSE), and the overall (F)-statistic.
- d) State the overall hypotheses and make the test decision using the critical value.
- e) Interpret the VIF values, Durbin-Watson value, residual spread and Q-Q plot.
- f) Give a balanced conclusion about what is satisfactory and what should be investigated before using the model.